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ORIGINAL ARTICLE

Automatic berthing control under wind disturbances and its implementation in an embedded system

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Abstract
This paper proposes a practical algorithm and its implementation of automatic berthing in wind disturbances environments. Berthing operation is one of the most complex tasks for seafarers. Automation of the highly burdensome berthing maneuver can be used to assist seafarers. In this study, we analyze the effect of wind disturbance on path following and propose a new path following control algorithm using a 2-DoF controller that introduces a feed-forward control. We also propose a method to reduce path deviation by introducing a runway in path planning. The effectiveness of the proposed method is verified by numerical simulations and shipboard tests using an experimental ship, and the proposed method shows higher performance under wind disturbance than the previous methods. The algorithm verified in this study is implemented as a PLC system, AutoBerth PLC. A PLC is an industrial computer, stable in operation. Using the AutoBerth PLC, we conducted an experiment of automatic berthing by an actual ship and confirmed that the system can be controlled faster and more stable than when controlled by a laptop. To realize a practical system, an alert function for the ODD, an alert function for fallback response regarding self-diagnosis of PLCs and communication monitoring functions were implemented as required functions in a maritime autonomous surface ship (MASS), and these functions were verified on the experimental ship.

Keywords Automatic berthing · Wind disturbance · Programmable logic controller · Autonomous ship · Operational design domain

1 Introduction

1.1 Background and contribution

Berthing operation is a formidable task for crews and there is a demand for assistance in this operation. In addition to the above, with the recent discussion and active research and development of Maritime Autonomous Surface Ship (MASS) [1], automatic berthing systems have become an essential component of a MASS. Along with improvements in algorithms, safety measures and a user interface of the system are important in developing a practical automatic control system including berthing control. There are two major elements for realizing a practical automated berthing system: performance and safety. The performance of the algorithm includes improvement of robustness against disturbances. The safety covers safety measures and user interface/user experience (UI/UX). The latter is essential to ensure that the system is easy to use and provides information clearly, ultimately allowing the system to operate safely, but has not been discussed much in past research on automatic berthing control.

The authors studied automatic berthing control by applying the path following algorithm in the previous study [2]. In the previous study, the authors proposed an algorithm for berthing control and a method of correcting the approach angle so that the operator can easily judge the status of automatic control. The previous method did not have a controller to handle a disturbance, especially wind. This study proposes a simple method for automatic berthing control under a wind disturbance, not compromising the real-time performance and effect of the control method proposed in the previous paper.

The practical system requires an operational design domain (ODD). An ODD defines the operational conditions

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of the automated system and includes environmental, geographical, time-of-day, and traffic environment restrictions. This concept comes from the terminology of driving automation of cars. It is applied in level 3 (high driving automation) and 4 (conditional driving automation) automation except for level 5 (full driving automation) [3]. In this paper, we implemented an alert system, basing settings on the ODD for the control start position and a wind disturbance. The developed system assumes the MASS is manned for fallback. The alerts assist us in determining whether or not to continue automatic berthing. The onboard GUI will provide clear information for instantaneous determination of the situation under control. The GUI is based on the UI of the automatic berthing software that has been developed for laptops as shown in Fig. 1.

We present the practical automatic berthing system with anti-wind disturbance control and give an implementation of the automatic berthing system. The MASS system, including automatic berthing control, is required to indicate its limitations to operators and to constantly monitor the system's integrity and whether the system is within the ODD. The algorithm and system proposed in this study will be implemented in a Programmable Logic Controller (PLC). We also implemented alerts for ODD and system self-diagnostics in a system named AutoBerth PLC. PLCs are widely used in commercial ships to collect sensor data of various onboard equipment. PLCs are industrial-grade computers and stable and reliable hardware compared with consumer personal computers. We demonstrate that the proposed algorithm of anti-wind disturbance control is capable of stable berthing under wind disturbances in actual sea experiments. After this experimental validation, we evaluated the AutoBerth PLC compared with the results using the laptop. The system achieved more stable control compared to the system on the laptop.

1.2 Related work

There are many related works on automatic berthing under wind disturbances. Ahmed et al. [4] proposed a controller consisting of an artificial neural net (ANN) and a proportional-differential (PD) controller. They have verified the controller performance in a wind environment by numerical simulations of ESSO Osaka in the model scale. The experimental results showed that control by neural networks alone cannot provide accurate control against wind effects and that PD control is indispensable. Shuai et al. [5] also provided results of numerical simulation using an ANN-based controller under static and variable wind speed conditions. Their study was validated only in the wind direction of away from the dock. Under static wind disturbance, the ship approached the dock on a stable straight trajectory, but under variable wind conditions, the ship drifted to the downwind side. Mizuno and Kuboshima [6] utilized a three-layered recurrent neural network (RNN) for automatic berthing control. They carried out experiments using an actual ship of 425GT equipped with a controllable pitch propeller (CPP). Mizuno et al. [7] proposed the quasi-real-time optimal scheme for automatic berthing control. This scheme consists of a nonlinear path optimization based on the multiple shooting algorithm and model predictive control (MPC). This method is also verified in experiments using an actual ship. Using this method, the time required to calculate the optimal path to a berthing point is from 3 to 5 min using this method, and the ship was carried away under wind disturbance before control begins. In a similar study, Martinsen et al. [8] reported the automatic berthing for an autonomous surface vehicles

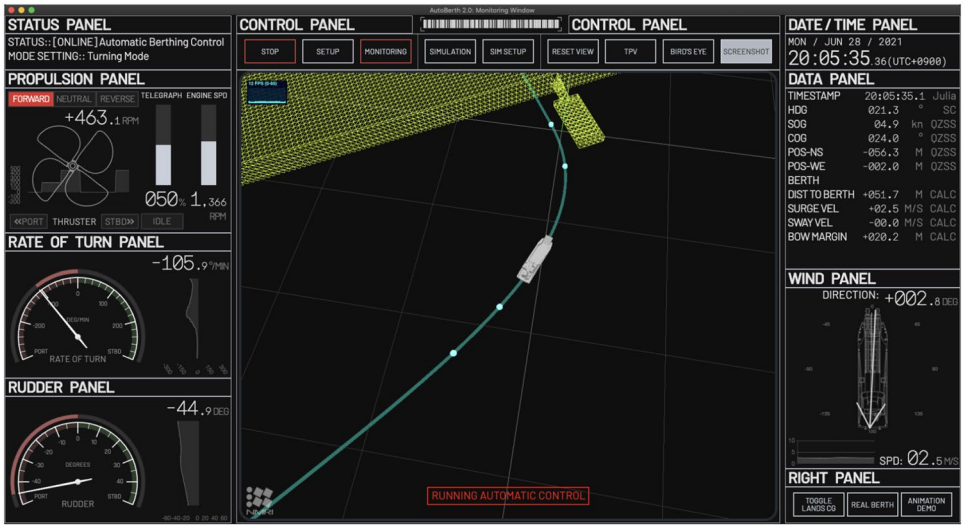

Fig. 1 Graphical user interface of automatic berthing software running on a laptop

(ASVs). They formulated a nonlinear optimal control problem and applied it to a trajectory-tracking dynamic positioning controller and verified using the small ASV in calm winds between 1 [m/s] and 3 [m/s]. Other results of the application of nonlinear optimization methods to automatic berthing can be seen in works of Maki et al. [9] and Akimoto et al. [10]. Okuyama et al. [11] proposed a minimum time harbor operation system for coastal vessels and applied their method to berthing operation using an actual ship at Ube port in Japan. Minimum time maneuvering solutions were obtained using the sequential conjugate gradient-restoration algorithm (SCGRA), which was solved by Ohtsu et al. [12] in 1995. The proposed method is applied to harbor operation before approaching maneuvers to the berth in the experiment. Their method could account for wind disturbance, but in both experiments, the conditions were windless. In all of these studies, ships with a CPP were used. Since CPP can continuously control the blade angle of the propeller, it is easier to control the ship's speed than with a fixed pitch propeller. Few studies are using an actual ship with a fixed propeller except [2]. Han et al. [13] proposed a motion planning approach using an extended dynamic window approach (EDWA) for automatic berthing control for an underactuated unmanned surface vehicle (USV). This approach considers dynamic constraints of the USV, obstacles, and effects of wind loads and provides an optimal trajectory. They conducted case studies with numerical simulations changing wind speed and direction. Piao et al. conducted experiments with a ship model under wind disturbances with the wind-making machine [14]. They applied the line of sight (LOS) algorithm for path planning and the active disturbance rejection control (ADRC) for control under wind disturbances. The experiment was conducted under 1.5 [m] wind disturbance conditions, and the PID controller failed to arrive at the target berthing area. Still, the ADRC controller could control the model ship to the target berthing area. Shimizu et al. [15] utilized supervised learning (SL) and reinforcement learning (RL) to perform automatic berthing control. This approach takes into account the safety risk of collision with the berth and they evaluated the method in numerical simulation at model-scale with wind disturbances. Liu et al. [16] proposed the method with the barrier Lyapunov function (BLF) for automatic berthing control.

This paper also provides a method of implementing the automated navigation system, using our automatic berthing system as an example. In Japan, several demonstrations of automatic operation systems, including automatic berthing control, were conducted on actual vessels under the MEGURI2040 project [17, 18]. However, these studies did not address the system monitoring or alert management for berthing function. As a related work, Yong et al. [19] proposed the automatic docking system for the mobile harbor. Their docking system has a passive mode for the normal state of operation and the emergency mode, which allows the ship and the mobile harbor for emergencies. As an example of actual social implementation, we would like to mention the standard of Track Control Systems (TCS) [20]. It defines the minimum operational and performance requirements and tests conforming to performance standards adopted by International Maritime Organization (IMO) for TCS. It requires that the system has an appropriate alarm for emergency or failure of the system in the standard.

2 Subject ship and control system

2.1 Subject ship and onboard control system

The small ship is chosen as a target of this research because it is expected to be significantly affected by disturbances such as wind at a low speed than larger vessels. In addition, to apply to a wide range of existing ships in operation, a ship with one propeller and one rudder is assumed as a specification of the target ship. For these reasons, we utilized the experimental ship "Shinpo" managed by the National Maritime Research Institute shown in Fig. 2, which meets these conditions. Table 1 shows the principal particulars of Shinpo at the time of experiments. Shinpo is equipped with an MAU-type propeller and a bow thruster.

An onboard control system was constructed to perform automatic berthing control [2]. The architecture of the onboard control system is shown in Fig. 3. The control system consists of programmable logic controllers (PLCs), which are reliable and scalable, that can control various actuators of the ship. All measured sensor values are aggregated into the onboard PLC control system, recorded and displayed on the touch display connected to the control system. The hydraulic steering system and the remote control system of the main engine can be monitored and operated by the onboard PLC control system, which can operate the rudder,

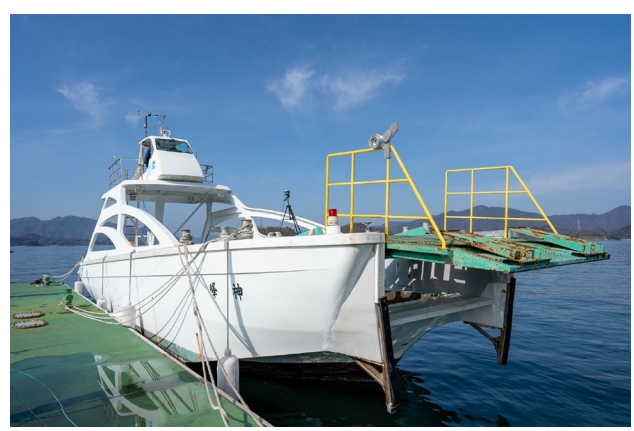

Fig. 2 Experimental ship "Shinpo"

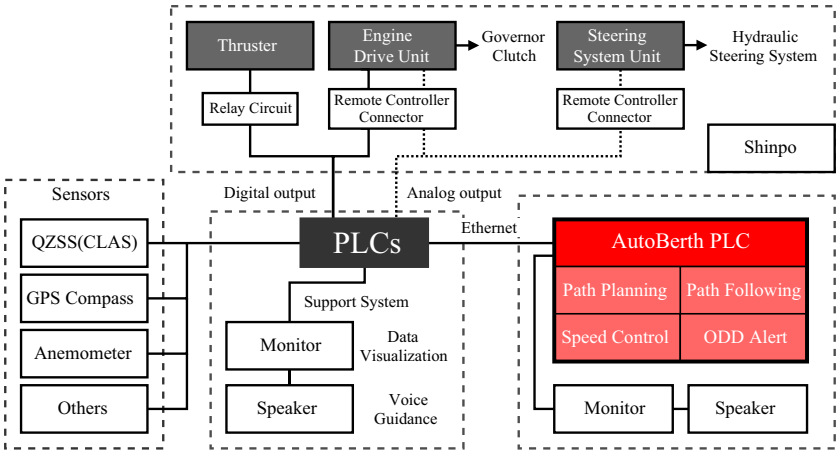

Fig. 3 Architecture of onboard control system and AutoBerth PLC

Table 1 Principal particulars of the experimental ship "Shinpo"

Subjects	Value
Ship length overall, L_{oa} [m]	16.5
Ship length between perpendiculars, L_{pp} [m]	14.9
Ship breadth (1.0WL), B [m]	4.38
Ship draft, d_m [m]	0.502
Trim, τ [m]	0.348
Diameter of propeller, D [m]	1.0
Propeller pitch @ 0.7R, P [m]	0.775
Side area of rudder, A_R [m^2]	0.780
Height of rudder, H_R [m]	1.05
Mass, m [kg]	21.53×10^3
Moment of inertia, I_{zG} [kg·m^2]	3.044×10^2
Longitudinal coordinate of center of gravity, x_G [m]	− 1.260
Longitudinal projected area, A_F [m^2]	14.40
Lateral projected area, A_L [m^2]	41.52

governors (corresponding to engine telegraph command), the clutch, and the bow thruster used in berthing operations. The automatic berthing algorithm is implemented on a laptop computer in the previous study [2], which can be connected to the onboard control system. In this study, we first used a laptop computer instead of "AutoBerthPLC" to verify the newly proposed algorithm. The results are presented in the Sects. 5.2 and 5.3. After that, the validated algorithm is implemented in the PLC system as described in the next section and evaluated using the experimental ship in Sect. 5.4.

2.2 AutoBerth PLC

In this section, we show an example of a practical implementation of the automated system for berthing control. The proposed algorithm validated in the previous section was implemented as an automated system using a programmable logic controller (PLC). We named this system AutoBerth PLC. PLC is an industrial computer used as a real-time controller. PLCs are used in typical commercial ships as processing devices for onboard communication of sensor data, data management, and analysis, such as mileage monitors and main switchboards. In addition, on the experimental ship, the PLCs are used in the onboard control systems for collecting sensor data, analysis, logging, voice support of berthing operation and waypoint navigation, and control [2]. Figure 4 shows the AutoBerth PLC system implemented in this study. The PLC is installed in a waterproof case, and a touch display on the case can display information and issue voice guidance and alerts. To make the case waterproof, power and Ethernet cables are routed outside the case through waterproof plastic cable glands.

For programming software running on PLCs, the five programming language sets are defined in IEC61131–3 [21]. Programmers of PLCs basically use the ladder diagram (LD) to describe processes. Figure 5 shows an example of the ladder diagram in AutoBerth PLC. The basic functions of the proposed system, such as self-holding circuits, timer processes and triggering alarms, were implemented using the ladder diagram. Complicated processes including functions and conditional execution such as IF-ELSE-THEN clause in the automatic berthing control code are implemented in the structured text (ST) language. Calculations of the path planning and the automatic berthing control of the proposed system were implemented by the ST language. The system architecture of AutoBerth PLC and the onboard control system is shown in Fig. 3. The AutoBerth PLC can display sensor data received from the onboard control system in an easily understood manner. A screenshot of the actual system GUI is shown in Fig. 6. The system also has an option menu for ODD alerts settings, emergency stops, and status monitoring, as described in this section (Fig. 7).

In general, MASS automation systems should be designed to operate within a specific ODD. Specifically, the system should have a function to monitor whether the system is within the ODD or alert and fallback response if the system moves out of the ODD. The fallback response

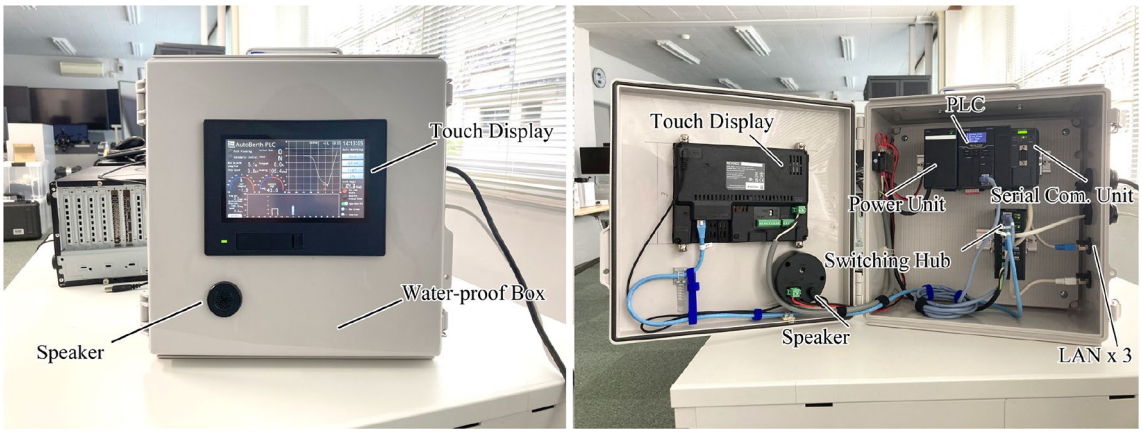

Fig. 4 Automatic berthing (AutoBerth) PLC system

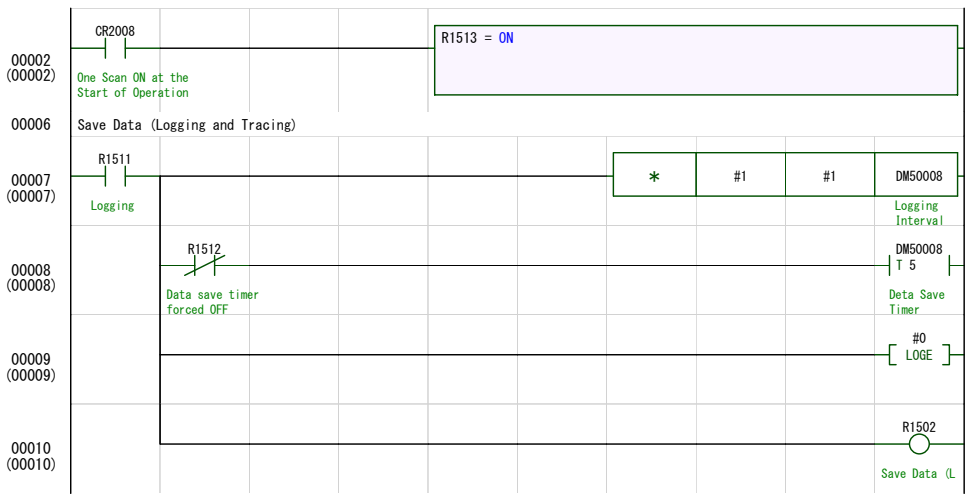

Fig. 5 Example of the rudder diagram in AutoBerth PLC (translated by the authors). In the diagram, "R" is a relay, "DM" is a data memory, "T" is a timer and "CR" is a special relay named the control relay. For example, lines 00007 to 00010 are executed when R1511 is ON in this implementation

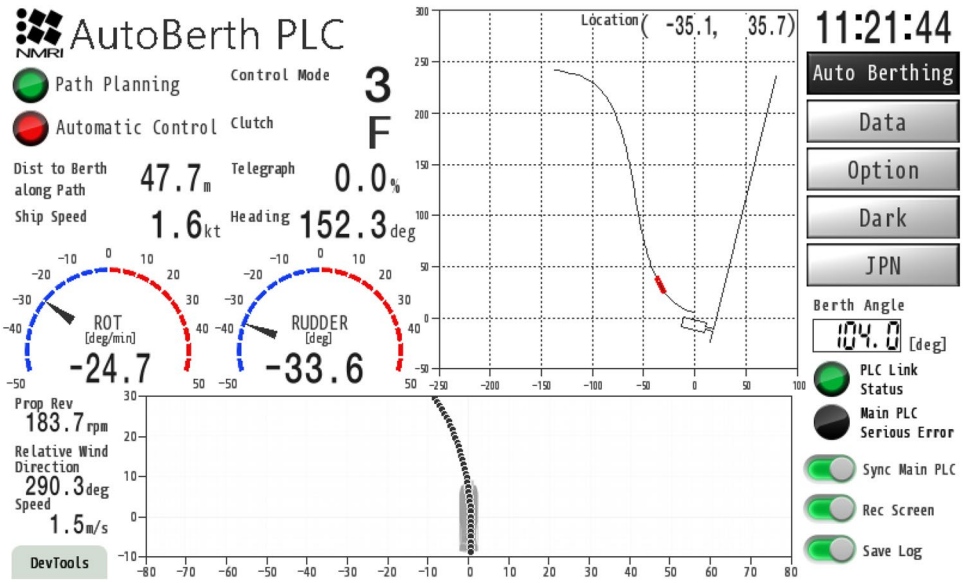

Fig. 6 Main graphical user interface of AutoBerth PLC

Fig. 7 Option menu of Auto-Berth PLC

includes a minimum risk maneuver that automatically transitions to the minimum risk condition defined at the time of design, using normally operating equipment when the system detects a fault condition of the system. Such functions are used at the automation level such as unmanned ships, remotely operated ships, or the manned ships with no one in the bridge at any time. In this study, we assumed a situation where the ship is manned onboard and the system is to perform automatic control, and we implemented an alerting function to provide fallback responses, such as alerting by the system and handing over to the onboard operator. The human operator is supposed to constantly monitor the control status during automatic berthing, and is alerted when the system is out of the ODD or when some abnormal condition occurs in the system. Some of the conditions covered by the ODD change rapidly during operation, such as wind conditions. For such conditions, the system should be designed to have a fixed measurement period and not be affected by changes in instantaneous values. In this study, we implemented a function to set ODD for the control start position and wind speed. This ensures that the control start position is within a predefined range at the start of the control and that the average and maximum true wind speeds over a 5-min period are below the predefined criteria.

One of important functions of the PLC is the self-diagnostic function. This constantly monitors the status of a PLC and stores the PLC status in a dedicated memory. In the proposed system, the main onboard control system and AutoBerth PLC communicate and operate via Ethernet. In this study, we implemented a function that monitors the communication status of each other's system and a function that monitors the self-diagnostic results of each other's system and alerts when an error occurs when disconnected. This function, for example, alerts operators using voice guidance and alarm when an Ethernet cable is disconnected during running, thereby notifying them of an abnormal communication status and prompting them to take action. Abnormal conditions are classified into two types: minor errors, such as calculation errors, and major errors. In particular, we implemented a function to alert operators of major errors that may render the system inoperable. Such an alert function for fallback response in the AutoBerth PLC was installed on the experimental ship and confirmed to be functional.

3 Simulation model

The mathematical model of maneuvering motion of the subject ship is described in this section. The model is basically the same as the previous study [2]. This model is utilized for the development and evaluation of the automatic berthing algorithm for Shinpo under various wind conditions. Unlike general maneuvering motions such as turning motion, a model that corresponds to changes in the ship's speed, including propeller reverse rotation, is required for berthing operation. In addition, as mentioned in the paragraph on ship-mounted sensors, high-precision motion control is required for berthing control, so the accuracy of motion prediction in the simulation must be sufficiently high. The dynamic model was based on the Mathematical Modeling Group (MMG) model [22], which can predict the maneuvering motion of a ship with high accuracy. Hydrodynamic coefficients and other parameters of the model were determined based on the results of sea trials using Shinpo. In this section, we first describe the MMG-based motion equations of the maneuvering motion. Then, we give a brief

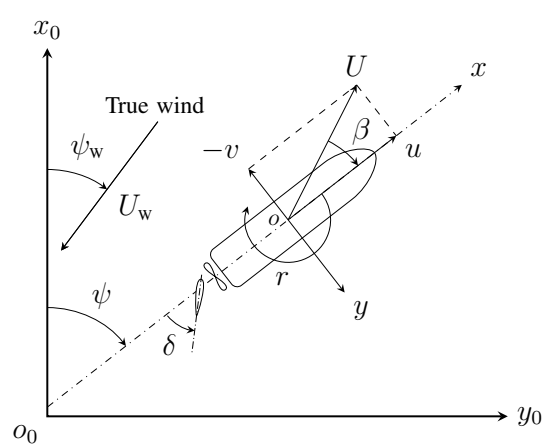

Fig. 8 Coordinate systems of ship motion and wind

description of the procedure to adjust the parameters of the dynamic model and wind simulation. Waves and tidal currents are not included in this study because they are hardly observed in the sea area around the floating pier.

3.1 Mathematical model of ship maneuvering motion

We use the mathematical model of maneuvering motion based on the MMG model to evaluate the response against the algorithm under disturbance, especially the effect of wind for ship motion at low speed. Figure 8 shows the coordinate systems used in this paper in order to describe motion equations. The notation for each symbol is in accordance with the paper [22].

$$\left.\begin{aligned} X &= X_H + X_R + X_P + X_A \\ Y &= Y_H + Y_R + Y_P + Y_A \\ N &= N_H + N_R + N_P + N_A \end{aligned}\right\}. \tag{2}$$

Here, subscript H, R, P and A means hull, rudder, propeller and air (wind), respectively. Force due to the bow thruster is omitted, because the bow thruster is not used in this study.

For the calculation of the hydrodynamic force acting on the ship's hull, we use the following model proposed by Yoshimura et al. [26] which was based on cross-flow drag theory. This model can express the hydrodynamic force in the transverse motion, even as where the drift angle are much developed at low speed. From a practical viewpoint, construction of this model requires less efforts due to a smaller number of coefficients compared with the standard MMG model such as [22].

$$\left.\begin{aligned} X_H &= \frac{1}{2}\rho L d\left[\left\{X'_{0F} + (X'_{0A} - X'_{0F})\frac{|\beta|}{\pi}\right\}uU + \left(m'_y + X'_{vr}\right)L\cdot vr\right] \\ Y_H &= \frac{1}{2}\rho L d\left[Y'_v v|u| + Y'_r L\cdot ur - \frac{C_D}{L}\int_{-L/2}^{L/2}|v + C_{rY}rx|(v + C_{rY}rx)dx\right] \\ N_H &= \frac{1}{2}\rho L^2 d\left[N'_v v|u| + N'_r L\cdot |u|r - \frac{C_D}{L^2}\int_{-L/2}^{-L/2}|v + C_{rN}rx|(v + C_{rN}rx)xdx\right] \end{aligned}\right\}, \tag{3}$$

The space-fixed coordinate system is denoted by $o_0 - x_0y_0z_0$ as shown in the Fig. 8. And $o - xyz$ is the ship-fixed coordinate system, where o is taken on the midship of the ship. u, v and r are the surge velocity, the lateral velocity at center of midship, and the yaw rate, respectively. The heading angle is denoted by ψ. The drift angle at midship position β is defined as $\beta = \tan^{-1}(-v/u)$, and the resultant velocity is represented as U, where the speed is $U = \sqrt{u^2 + v^2}$. The fundamental equations of ship maneuvering motion in 3-DOF is represented as Eq. (1).

$$\left.\begin{aligned} (m + m_x)\dot{u} - (m + m_y)vr - x_G m r^2 &= X \\ (m + m_y)\dot{v} + (m + m_x)ur + x_G m\dot{r} &= Y \\ (I_{zG} + x_G^2 m + J_z)\dot{r} + x_G m(\dot{v} + ur) &= N_m \end{aligned}\right\}. \tag{1}$$

Here, the added masses of surge and sway motion and the added moment of inertia of yaw motion are denoted by m_x, m_y and I_z, respectively. Added masses and the added moment of inertia are determined using the multiple regression formulae of Motora's chart [23–25].

Forces X, Y and moment N_m acting on a ship's hull around the midship can be described separating into the following components from the viewpoint of the physical meaning according to the manners of the MMG standard method.

where X'_{0F} and X'_{0A} are resistance coefficients of ahead and astern, respectively. Here, X'_{vr}, Y'_v, Y'_r, N'_v and N'_r are hydrodynamic derivatives. C_D is the cross-flow drag coefficient, C_{rY} and C_{rN} are correction factor for lateral force and yaw moment, respectively. In this study, the resistance coefficient for ahead X'_{0F} is determined using the result of the speed trial test in full scale, where propeller rotation speed is 3.1 rps. The coefficient X'_{0A} is determined from the database of the tank test results for the astern resistance test.

Hydrodynamic force due to rudder and propeller with forward and reverse rotation is expressed as the model by Kitagawa et al. [27, 28] including the unbalanced hydrodynamic force due to propeller reverse rotation.

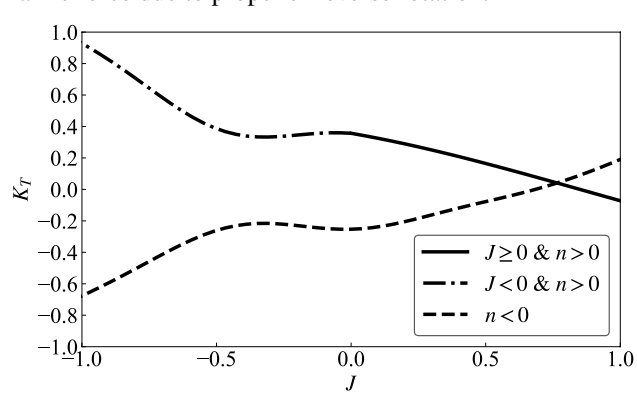

Fig. 9 Estimated K_T-J curves based on the estimate formula of MAU series propellers for the 1st quadrant and the database of B-series propellers for 2–4th quadrants

$$\left.\begin{aligned} X_R &= -(1-t_R)F_N \sin\delta \\ Y_R &= -(1+a_H)F_N \cos\delta \\ N_R &= -(x_R + a_H x_H)F_N \cos\delta \end{aligned}\right\}. \tag{4}$$

$$\left.\begin{aligned} X_P &= (1-t_P)\rho n^2 D^4 \cdot K_T(J) \\ Y_P &= \begin{cases} 0, \text{ for } n \geq 0 \\ \rho n^2 D^4 \cdot Y_P^*, \text{ for } n < 0 \end{cases} \\ N_P &= \begin{cases} 0, \text{ for } n \geq 0 \\ \rho n^2 D^4 L(x'_{PRR} Y_P^*), \text{ for } n < 0 \end{cases} \end{aligned}\right\}, \tag{5}$$

where $(1-t_R)$ is the steering resistance deduction factor, a_H is the rudder force increase factor, and x_H is the longitudinal coordinate of acting point of the additional lateral force. The thrust deduction factor $(1-t_P)$ is different depending on the sign of propeller revolution n. The propeller thrust open water characteristic $K_T(J)$ for each quadrant shown in Fig. 9 are obtained from the estimation formula of MAU propellers for 1st quadrant and the database of B-series propellers for 2–4th quadrant. The propeller advanced ratio J is represented as

$$J = \frac{u(1-w_P)}{nD}, \tag{6}$$

where D is the propeller diameter and n is the propeller revolution. The wake coefficient at propeller position is denoted by w_P. Assuming that ship's speed is controlled only by switching the clutch in the idle state of the main engine, the propeller revolution n corresponds to 3.1, 0, and −3.1 rps for forward, neutral, and reverse of the clutch based on the actual ship measurement. In the simulations, the time delay of switching the clutch was ignored. When the clutch condition is neutral, the propeller revolution is 0, so the calculation of J is omitted. Then, the propeller thrust $T = X_P/(1-t_P)$ is treated as 0, and the longitudinal inflow velocity to the rudder $u_R = (1-w_R)u$ is computed based on the original definition of u_R [22, 29]. Thrust deduction factor t_P for $n < 0$ can be computed as follows:

$$1 - t_P = C_{tp1} \cdot J_P + C_{tp0}, \tag{7}$$

where $J_P := \frac{u}{nP}$ is the so-called apparent advanced ratio used in Kitagawa's model. C_{tp0} and C_{tp1} can be changed depending on J_P as shown in Table 3, whose values are identified by the database of captive model tests with propeller reverse rotation. The rudder normal force F_N is expressed as

$$F_N = \frac{1}{2}\rho A_R U_R^2 C_N(\alpha_R). \tag{8}$$

Here, the resultant rudder velocity U_R and its effective inflow angle to rudder α_R is expressed as follows:

$$\left.\begin{aligned} U_R &= \sqrt{u_R^2 + v_R^2} \\ \alpha_R &= \delta - \delta_0 - \tan^{-1}\left(\frac{v_R}{u_R}\right) \end{aligned}\right\}, \tag{9}$$

where δ_0 is a virtual zero rudder angle for the rudder normal force. The rudder normal force coefficient C_N is expressed as follows:

$$C_N(\alpha_R) = \begin{cases} C_{N0} \sin\alpha_R, \text{ for } 0 \leq |\alpha_R| < |\alpha_{Rstl}| \\ \text{or } (\pi - |\alpha_{Rstl}|) \leq |\alpha_R| \leq \pi \\ C_{N0} \sin(\text{sgn}(\alpha_R) \cdot \alpha_{Rstl}), \\ \text{for } |\alpha_{Rstl}| < |\alpha_R| < (\pi - |\alpha_{Rstl}|) \end{cases}, \tag{10}$$

where C_{N0} is the gradient coefficient of the rudder normal force, which is determined using the Fujii's formula [30]. The stall angle of the normal rudder force is denoted by α_{Rstl}. Then, the inflow velocity to the rudder u_R is defined as follows:

$$u_R = \begin{cases} u(1-w_R)\sqrt{\eta \cos^2\alpha_S \cdot \left\{1 + k_x \frac{1-w_P}{1-w_R}\left(\sqrt{1+\frac{8K_T}{\pi J^2}} - 1\right)\right\}^2 + (1-\eta)}, \text{ for } n \geq 0 \\ \text{sgn}(u_{RPRsq})\sqrt{(|u_{RPRsq}|)}, \text{ for } n < 0 \end{cases}, \tag{11}$$

where

$$\left.\begin{aligned} u_{RPRsq} &= \eta \cdot u_{PRR1}|u_{PRR1}| + (1-\eta) \cdot u_{PRR2}|u_{PRR2}| + C_{PRR} \cdot u|u| \\ u_{PRR1} &= u(1-w_R) + \cos\alpha_S \cdot nD_P k_{xPR}\sqrt{\frac{8|K_T|}{\pi}} \\ u_{PRR2} &= u(1-w_R) \end{aligned}\right\}. \tag{12}$$

The wake coefficient at rudder position in maneuvering motion is denoted by w_R. The term $\cos(\alpha_S)$ is a correction factor that takes into account the direction of the velocity increases due to propeller. In Eq. (12), k_x is an experimental constant which is called the increment ratio of propeller

Table 2 Hydrodynamic force coefficients

Parameter	Ahead ($u \geq 0$)	Astern ($u < 0$)
Y'_v	− 1.23	− 0.5
Y'_r	0.106	0.0476
N'_v	− 0.0791	− 0.0949
N'_r	− 0.134	− 0.0614
C_D	0.185	0.0834
C_{rY}	1.41	6.36
C_{rN}	3.34	1.53
Parameter	Forward ($n \geq 0$)	Reverse ($n < 0$)
Y^*_P	–	− 0.00924
x'_{PR}	–	− 0.278
δ_0 [deg]	− 2.47	5.0
x'_H	− 0.364	0.0

slipstream. Additionally, η is the ratio of the effective area of the rudder for propeller slipstream to the whole rudder area. Here, the lateral inflow velocity v_R is expressed as

$$v_R = -\gamma_R(v + l_R r), \tag{13}$$

where γ_R and l_R are flow straightening coefficients.

3.2 Wind forces

As an external force acting on the hull, wind force is treated here. Longitudinal and lateral forces, and yaw moment acting on the hull due to wind are expressed as follows:

$$\left.\begin{aligned} X_A &= \frac{1}{2}\rho_A A_F U_A^2 C_{AX}(\psi_A) \\ Y_A &= \frac{1}{2}\rho_A A_L U_A^2 C_{AY}(\psi_A) \\ N_A &= \frac{1}{2}\rho_A A_L L_{OA} U_A^2 C_{AN}(\psi_A) \end{aligned}\right\}, \tag{14}$$

where relationship between a true wind and an apparent wind is shown as follows:

$$\boldsymbol{U_A} = [u_A, v_A] \quad = \begin{bmatrix} -U_W \cos\psi_W - u\cos\psi + v\sin\psi \\ -U_W \sin\psi_W - u\sin\psi - v\cos\psi \end{bmatrix}^{\mathrm{T}}, \tag{15}$$

Table 3 Hydrodynamic coefficients and parameters for simulation

m'_x	0.0502	t_P ($n \geq 0$)	0.075	C_{tp0} ($J_P \leq -1$)	0.8294
m'_y	0.385	w_R	0.0	C_{tp1} ($J_P \leq -1$)	− 0.0071
J'_z	0.00235	α_{Rstl} [deg]	35.0	C_{tp0} ($-1 < J_P \leq J_{P0}$)	1.5320
X'_{0F}	− 0.049	α_S [deg]	10.0	C_{tp1} ($-1 < J_P \leq J_{P0}$)	0.6954
X'_{0A}	− 0.056	η	0.847	C_{tp0} ($J_{P0} < J_P$)	0.9246
X'_{vr}	0.055	C_{N0}	2.657	C_{tp1} ($J_{P0} < J_P$)	− 0.8628
w_P	0.0	x'_R	− 0.500	J_{P0}	− 0.3898

where U_W is the true wind speed, ψ_W is the true wind direction where 0° stands for a northerly wind, U_A is apparent wind speed and ψ_A is apparent wind direction where 0° stands for head wind. The direction of true wind is measured in degrees clockwise from due north and the heading of a ship. The direction of apparent wind is computed as $\psi_A = \tan^{-1}\left(v_A/u_A\right) - \psi_W$. The wind load coefficients C_{AX}, C_{AY} and C_{AN} were determined using the method proposed by Kitamura et al. [31]. The curves of wind load coefficients are shown in Fig. 13 in the next section. The wind conditions in simulation were generated in two settings: constant wind speed/direction and sampling using probability distributions. In the simulations using the probability distributions, the true wind speed was produced according to the Weibull distribution and the true wind direction according to the Gaussian distribution, as follows [32, 33].

$$f_{U_W}(U_W \mid a, \lambda) = \frac{a}{\lambda}\left(\frac{U_W}{\lambda}\right)^{a-1} e^{-(U_W/\lambda)^a}. \tag{16}$$

$$f_{\psi_W}(\psi_W \mid \overline{\psi}_W, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\psi_W - \overline{\psi}_W)}{2\sigma^2}\right). \tag{17}$$

Here, $\overline{U}_W$ is the mean true wind speed and $\overline{\psi}_W$ is the mean true wind direction. The shape parameter a in Eq. (16) was set to 2.0. The scale parameter λ is calculated as

$$\lambda = \frac{2\overline{U}_W}{\sqrt{\pi}}. \tag{18}$$

This formula is obtained from the formula of the expected value of the Weibull distribution, where $a = 2.0$: $\overline{U}_W := \mathbb{E}[U_W] = \lambda\Gamma(3/2)$. Here, $\Gamma(x)$ is the gamma function. The standard deviation σ is set to 30 degrees according to the measured wind direction in the test sea area.

3.3 Hydrodynamic coefficients and other parameters

The hydrodynamic coefficients and other parameters of the model were identified based on the measurements of sea trials. We carried out various tests such as turning tests with rudder angles of 20 and 40 degrees for ahead and

Table 4 Coefficients to representing force due to and rudder

Parameter	$n \geq 0$	$n < 0$
t_R	0.0897	0.642
a_H	0.169	− 0.696
x_H/L_{pp}	− 0.364	0.138
k_x	0.541	0.144
γ_R	0.452	1.0
l_R	− 0.931	− 0.5

astern, and Zigzag tests of ± 10 and ± 20 degrees only for ahead mainly with a propeller revolution of 3.1 rps, which is in the idle state of the main engine. Next, referring to various estimation formulae and databases [26, 27, 34, 35], initial parameters of the motion equations were set. However, since these estimated parameters are based on different ship types, the parameters were adjusted based on the measurement results of full-scale tests of Shinpo. Table 2 shows adjusted parameters in this procedure and other parameters of the model used in this research is shown in Tables 3 and 4.

4 Methods

4.1 Feedforward controller for wind disturbance: FFPPA algorithm

The proposed method in this study is based on the automatic berthing control algorithm with the path following algorithm of the previous study [2]. Figure 10 shows a scheme of the automatic berthing control developed in the previous study. The planned path is divided into four sections in the berthing control, and the control of the clutch system is switched between each section. The path planning uses an algorithm that automatically generates a berthing path based on four pieces of information: the initial ship's position, heading angle, berthing position, and berthing angle, using a cubic Bézier curve. The previous method in [2] utilized pure pursuit and autopilot (PPA) algorithm to follow a berthing path. The schematic diagram of the PPA algorithm is shown in Fig. 11. Pure pursuit performs tracking control by controlling the heading to aim slightly ahead of the path at a target point. A PD-based heading control system performs the heading control. The advantage of this control system is

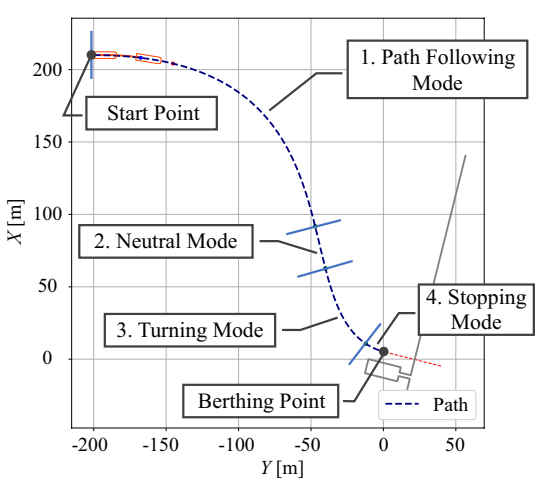

Fig. 10 Scheme of automatic berthing control

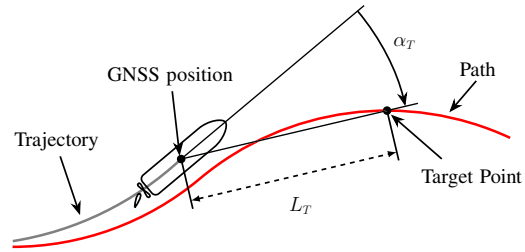

Fig. 11 Path following control using pure pursuit and autopilot (PPA)

that it requires only a few control parameters. In particular, the pure pursuit can be designed by simply adjusting the look ahead distance L_T which is the distance from the ship's position to the target point. The ship's speed is controlled by switching the clutch which is explained in detail in the previous study [2].

Figure 12 shows the deviation from the path as an effect of wind disturbance on the path following control. When a ship is subjected to wind from the side, the deviation from the path becomes larger. The accuracy of the path following control can be improved by reducing the path deviation caused by the wind. Therefore, we design a feedforward control system that determines the amount of steering to cancel the lateral component of the wind force. To simplify the discussion, consider the following equations of ship maneuvering motion with the center of gravity at the origin of the hull-fixed coordinate system.

$$\left.\begin{aligned}(m+m_x)\dot{u} &= (m+m_y)vr + X(u,v,r,\delta)\\ (m+m_y)\dot{v} &= -(m+m_x)ur + Y(u,v,r,\delta)\\ \left(I_{zG}+J_z\right)\dot{r} &= N(u,v,r,\delta)\end{aligned}\right\}, \tag{19}$$

where X, Y, N represent external forces and moment acting on the ship's hull around the center of gravity. The equation of motion contains many nonlinear terms including forces and moments acting on the hull. From this equation, the dynamics model for control is derived according to [36, 37]:

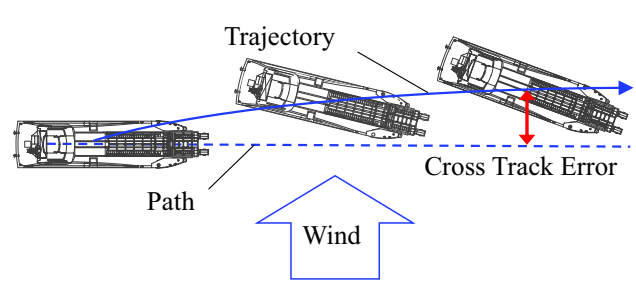

Fig. 12 Wind disturbance affecting path following control

$$
\begin{aligned}
(19) \Rightarrow & \begin{bmatrix} m+m_x & 0 & 0 \\ 0 & m+m_y & 0 \\ 0 & 0 & I_{zG}+J_z \end{bmatrix} \begin{bmatrix} \dot{u} \\ \dot{v} \\ \dot{r} \end{bmatrix} \\
= & \begin{bmatrix} X_l \\ -(m+m_x)ur+Y_l \\ N_l \end{bmatrix} \\
\Rightarrow & \begin{bmatrix} m+m_x & 0 & 0 \\ 0 & m+m_y & 0 \\ 0 & 0 & I_{zG}+J_z \end{bmatrix} \begin{bmatrix} \dot{u} \\ \dot{v} \\ \dot{r} \end{bmatrix} \\
= & \begin{bmatrix} X_u & 0 & 0 \\ 0 & Y_v & Y_r-(m+m_x)u \\ 0 & N_v & N_r \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix} \\
& + \begin{bmatrix} T_x & D_x \\ 0 & D_y \\ 0 & D_n \end{bmatrix} \begin{bmatrix} n_P \\ \delta \end{bmatrix} + \begin{bmatrix} X_A \\ Y_A \\ N_A \end{bmatrix},
\end{aligned} \tag{20}
$$

where X, Y and N represent linearized external forces and moment acting on the ship's hull around the center of gravity. X_u, Y_v, Y_r, N_v, N_r are hydrodynamic derivatives. The last term of Eq. (20) presents linearized force and moment by a propeller and a rudder. The terms of T_x, D_x, D_y, D_n are coefficients of the linearized force and moment. The sway motion equation in Eq. (20) is follows:

$$
\dot{v} = \frac{1}{(m+m_y)}\left[Y_r v + \left\{Y_r - (m+m_y)u\right\} + D_y\delta\right]. \tag{20'}
$$

Here, we define the lateral component of rudder force F_{dy} as $F_{dy} := D_y\delta$. Finally, the linearized lateral rudder force model for feedforward control is derived as follows:

$$
F_{dy}/(m+m_y) = D'_y\delta_{FF}, \tag{21}
$$

where D'_y is the normalized coefficient of the linearized model of lateral force induced by a rudder. The subscript FF indicates that the feedforward control law calculates the rudder angle. Assuming this control law, we analyzed the lateral forces acting on the hull due to steering using the mathematical model in the previous section. The analysis covers the motions corresponding to the forward, neutral, and reverse states shown in Table 5. For forward and neutral, the model is based on the motion state when the ship is moving straight ahead at the main engine idling speed.

Table 5 Settings for estimation of coefficients

Subjects	Forward	Neutral	Reverse
u [m/s]	2.072	2.072	0.0
v [m/s]	0.0	0.0	0.0
r [deg/s]	0.0	0.0	0.0
n_P [rps]	3.09	0.0	− 3.09

For reversing, the analysis is based on a zero ship speed condition because the clutch enters reversing only in the stop mode. The analysis results are shown in Table 6, where D_y is the coefficient calculated to satisfy Eq. (21). The lateral wind load acting on the hull is represented as the following model as stated in the previous section:

$$
Y_A = \frac{1}{2}\rho_A A_F U_A^2 C_{AY}(\psi_A). \tag{22}
$$

Figure 13 shows the wind load coefficients of the experimental ship. As a result, the rudder angle that cancels the lateral wind load can be obtained by solving $Y_A + F_{dy} = 0$ from Eq. (21) and Eq. (22).

By steering based on the obtained the rudder angle, the lateral component of the wind load can be canceled, but it is only an estimate, and other disturbance factors such as the moment by a wind load in the turning direction and currents, as well as errors in sensor values, may occur. Errors caused by these factors are collectively compensated for by the PPA algorithm, which has a feedback mechanism. The command value of the rudder angle δ^* is the sum of δ_{FF} calculated by the wind force compensation feedforward controller and the rudder angle δ_{PPA} calculated by the feedback controller using the PPA algorithm as follows:

$$
\begin{aligned}
\delta^* &= \delta_{PPA} + \delta_{FF} \\
&= K_P\alpha_T - K_D r - \frac{m+m_y}{2D'_y}\rho_A A_F U_A^2 C_{AY}(\psi_A),
\end{aligned} \tag{23}
$$

where K_P and K_D are the proportional and the derivative gain of the autopilot, α_T is the error of the target angle and r is the rate of turn. This is the feedforward, pure pursuit

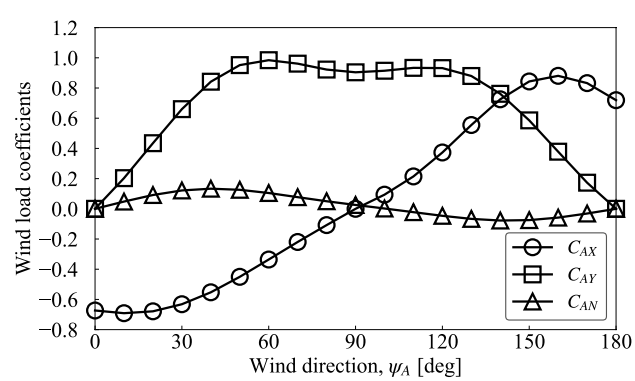

Fig. 13 Estimated wind load coefficients

Table 6 Estimated coefficients of linear lateral rudder force model

Subjects	Forward	Neutral	Reverse
D'_y	− 0.003272	− 0.002771	0.0

and autopilot (FFPPA) algorithm. This is a two-degree-of-freedom (2-DoF) controller. By expressing the relationship between the rudder angle and the wind load acting hull as a linear equation, the real-time performance of the conventional method can be maintained while controlling against wind disturbance. In this study, $K_P = 3.0$, $K_D = 1.0$ and the look ahead distance is $1.6L_{pp}$.

4.2 Path planning considering initial cross tracking error

The problem of outward drifting of the ship's trajectory during path following in the early stage of control is related to the situation in track control. The kick of the hull motion caused by steering during a turn becomes a problem when the ship moves from a leg consisting of straight lines to a curved track represented by an arc. The kick works in the opposite direction to that in which the ship turns, resulting in a delay in turning (Fig. 14). To avoid this problem, the start of the turn is usually set before the beginning of the arc path, as in Wheel Over Point (WOP), to reduce the delay in turning motion [38]. Regarding the previous path planning, the tangential direction of the path at the start position is parallel to the initial heading angle, but the path immediately turns toward the pier. In other words, the ship must already start turning at the start point. This may cause the bulging trajectory in the early stage of automatic berthing. We propose a solution to this problem by providing a straight runway section, as in the example of TCS on a curved track as shown in Fig. 15.

5 Experiments and evaluations

5.1 Environment and setting of experiments

This section provides results and evaluation of the proposed method in numerical simulation and full-scale experiments

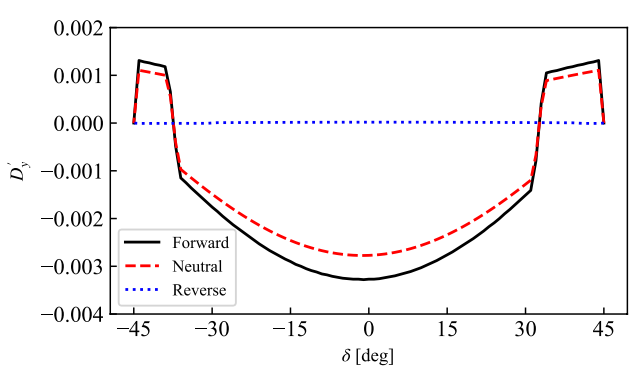

Fig. 14 Estimated coefficients of lateral rudder force

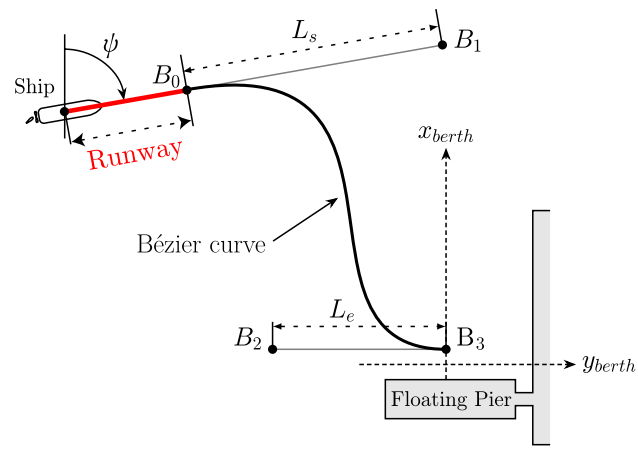

Fig. 15 Path planning using a cubic Bézier curve with a straight runway

using the experimental ship. The full-scale ship experiments were performed in the sea area at Innoshima Marina (Hiroshima, Japan) to validate the novel automatic berthing system as shown in Fig. 16. There is a floating pier, and the ship will berth with the starboard side. As shown in Fig. 16, there is a shallow sea area near the floating pier in the test sea area, and it is necessary to make a berthing path to avoid this.

The implementation of the whole automatic berthing control program is written in Julia [39] (controller) and JavaScript (GUI). The numerical simulation environments including the mathematical model and the wind simulation model is implemented in the automatic berthing control program. The real-time calculation of control inputs for full-scale experiments was performed using a laptop computer equipped with Core i7 (quad core, 2.8 GHz) and 16 GB RAM and running macOS Big Sur. The frequency of communication with the PLC system is set to 270 [ms] which is determined considering the stable and minimum interval for the serial communication between the laptop and the

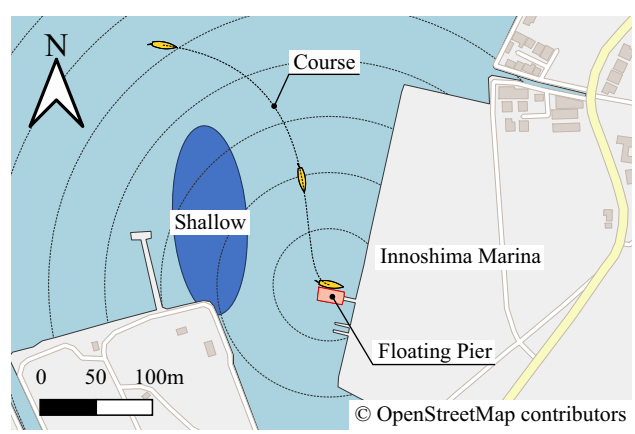

Fig. 16 Test area of automatic berthing experiments

onboard PLC system. For both numerical simulation and full-scale experiments, the terminal condition of automatic berthing control is defined as described in Fig. 17. The termination condition consists of the following three conditions: (a. the position condition) the midship position must be within the 8 [m] × 6 [m] rectangular area (gray in the figure), (b. the heading angle condition) the heading angle must be within from − 10 [deg] to +10 [deg] of the angle parallel to the floating pier (104 [deg]), and (c. the velocity condition) the ship's surge velocity must be less than 0.1 [m/s]. The condition of the midship position for the terminal condition was set so that the ship would be placed within the desired berthing area shown in the figure, and the distance of the berthing point (red dot) to the floating pier was set assuming a margin of 5 [m] which allows for the mooring rope to be taken on the pier from the ship. For the purposes of this study, automatic berthing is considered successful when these three termination conditions are satisfied.

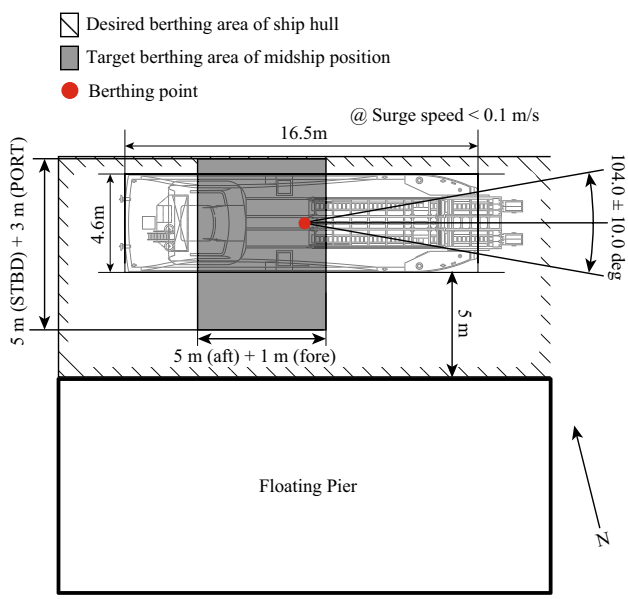

Fig. 17 Terminal condition of automatic berthing control

5.2 Numerical simulation

The results of numerical simulations of automatic berthing control using the FFPPA algorithm are shown in this section. For comparison, simulations were also performed using the PPA algorithm. First, simulations were performed under constant wind for various combinations of true wind speed and direction, and the results are shown in Fig. 18. The simulation results show that the PPA algorithm can handle all wind directions only at a true wind speed of 5 [m/s], while the FFPPA algorithm can handle all wind directions up to a true wind speed of 7 [m/s]. For true wind speeds greater than 7 [m/s], the FFPPA algorithm could not satisfy the set end conditions for some wind directions, but it resulted in more combinations of true wind speeds and directions that could be handled compared to the PPA algorithm. The same simulations were performed for variable winds. To qualitatively evaluate the effect of randomness in variable winds, we ran ten simulations each for each combination of true wind speed and wind direction and calculated the success rate. The results are shown in Fig 19. For the variable wind simulations, success rates by both algorithms were lower than for the steady wind simulations. As in the constant wind simulations, the FFPPA algorithm was successful in automatic berthing under more wind conditions than the PPA algorithm. The cross track error in the variable wind simulation is shown in Fig 20. The FFPPA algorithm is able to reduce cross track error as per the design aims of the controller described in the previous section. In particular, the difference between the PPA algorithm and the FFPPA algorithm is significant in the example of 135 and − 45 degrees wind directions, where the ship receives a crosswind when following the path. Conversely, in the 45 degrees and − 135 degrees examples, where the wind blows parallel to the path, the difference between the two algorithms is small, but there is a difference in the cross track error just before the floating pier, especially in the 45 degrees example.

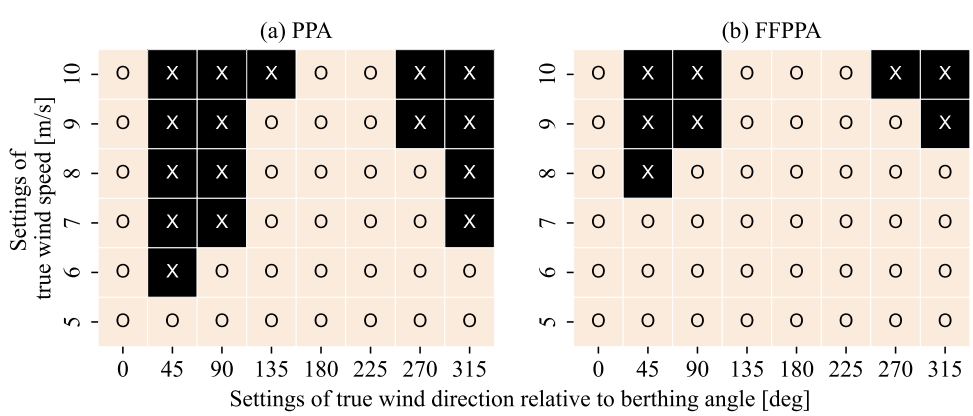

Fig. 18 Results of automatic berthing control under constant wind conditions using the PPA and the FFPPA algorithm in numerical simulation. The cream "O" cells represent successful berthing and the black "X" cells represent unsuccessful berthing. The initial ship's heading angle was 104 [deg] for all simulations

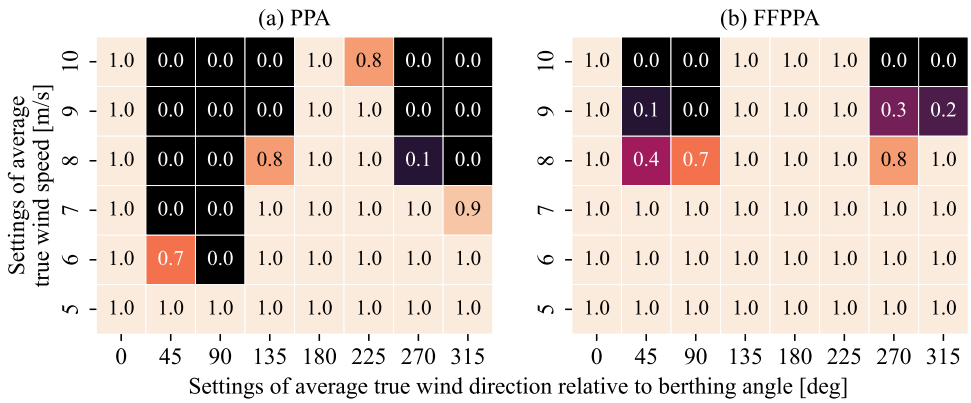

Fig. 19 Success rate of automatic berthing control under variable wind conditions using the PPA and the FFPPA algorithm in numerical simulation. Each case was simulated ten times to calculate the success rate. The initial ship's heading angle was 104 [deg] for all simulations. The winds were simulated using Eq. (16) for the speed and Eq. (17) for the direction of a true wind. The average true wind speed was determined as the scale parameter of the Weibull distribution calculated using Eq.(18). The variance of the wind direction was set at 30 deg based on the observation around the test sea area

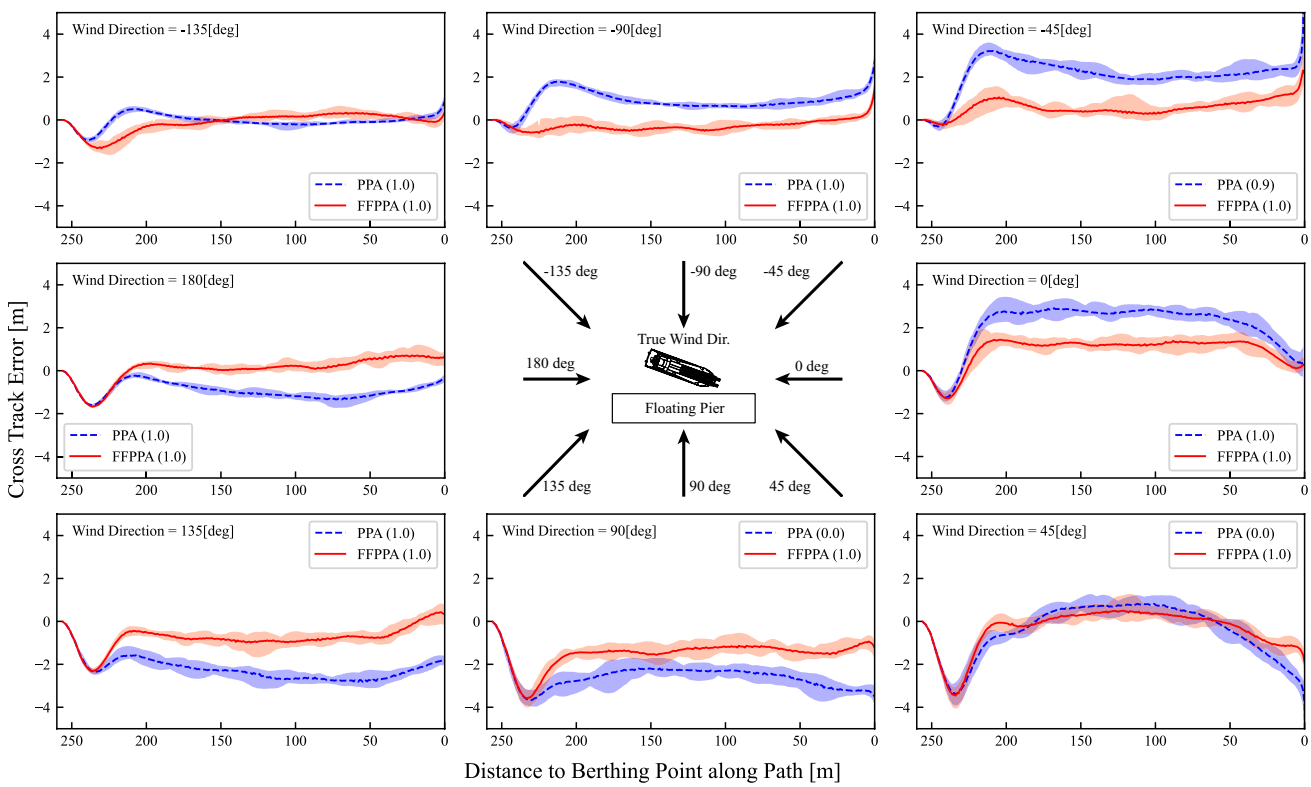

Fig. 20 Cross track errors during automatic berthing control under variable winds at the average wind speed setting of 7 [m/s] in eight directions using PPA and the FFPPA algorithm in numerical simulation. The suffix of each legend represents the results of berthing control that (O) is success and (X) is a failure of automatic berthing. The winds were simulated using Eq. (16) for the speed and Eq. (17) for the direction of a true wind, where $\lambda = 14/\sqrt{\pi}$ [m/s] for true wind speed, $\sigma = \sqrt{30}$ [deg] for true wind direction. Each case was simulated ten times. Each line shows mean value and an area surrounding each line is filled between minimum and maximum values computed at every 1.0 m of the distance of the berthing point on a path. The initial ship's heading angle was 104 [deg] for all simulations. The cross track error is negative if the ship is left of the intended path

Simulations were conducted for three wind conditions (no wind, headwind, and tailwind) at different runway lengths to investigate the effect of setting the runway on the suppression of cross track errors immediately after the start of control. The simulation results are shown in Fig. 21. For reference, the look ahead distance for the pure pursuit was 23.84 [m], which is $0.6 \cdot L_{pp}$. In all cases, regardless of the lengths of the runway, the initial cross-track errors were clearly suppressed compared to the case without the runway. The case of 20 [m] runway shows the smallest overshoot and fastest settling and was used in the full-scale experiments. The results using the runway in the full-scale experiments are shown in Sect. 5.4.

5.3 Full-scale experiments at sea

In this section, we present the results of full-scale experiments using the experimental ship Shinpo. Figure 22 shows the experimental results of the proposed wind disturbance compensation type path-following control. As shown in Fig. 22, this experiment was conducted under an average true wind speed of 6.38 [m/s]. The initial deviation was gradually reduced, and the deviation was almost eliminated by the time before the turning mode, which requires control accuracy at a distance of 50 [m] from the floating pier. The previous results without feedforward control for wind disturbances are shown in Fig. 23 for comparison. Although the average true wind speed was 6.34 [m/s] for this experiment,

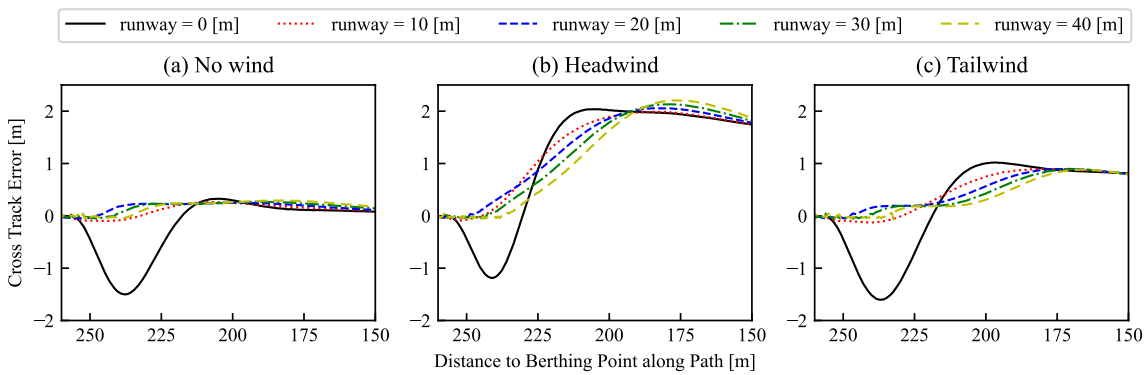

Fig. 21 Cross track error of early stage of automatic berthing control with various straight runway distance of paths in numerical simulation. In all cases except (**a**), a constant wind was simulated and the true wind speed was 8.0 [m/s], the direction was 104 [deg] for (**b**) and 284 (=104+180) [deg] for (**c**). The initial ship's heading angle was 104 [deg] for all simulations. The cross track error is negative if the ship is left of the intended path

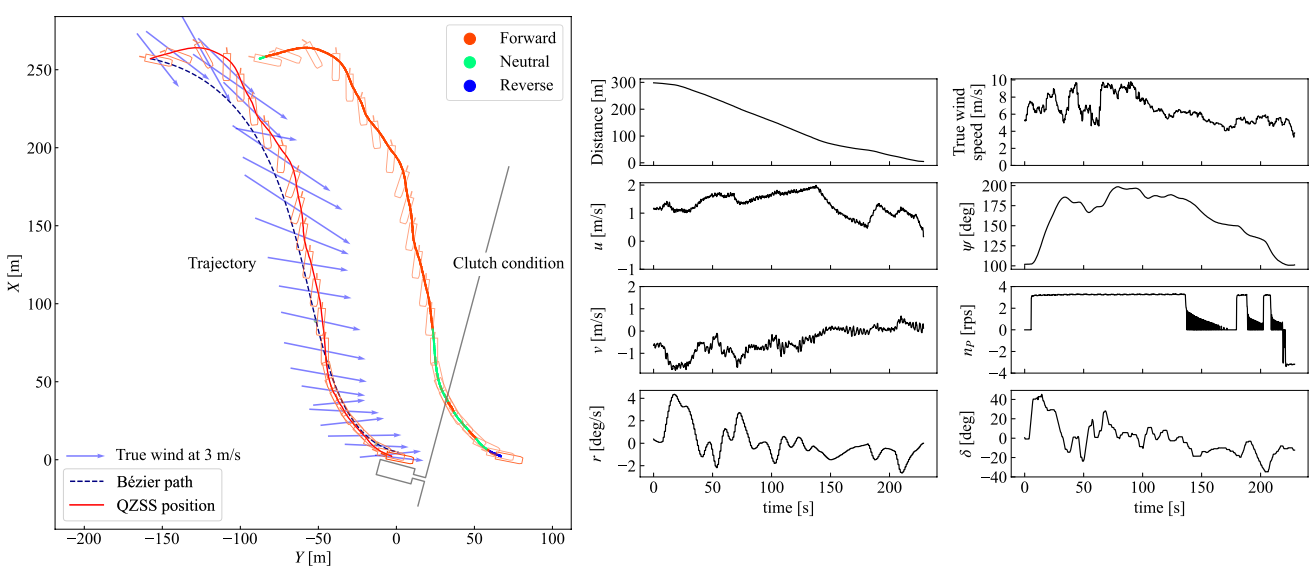

Fig. 22 Trajectory and measured values of automatic berthing control using the proposed method (FFPPA) in wind disturbance environment

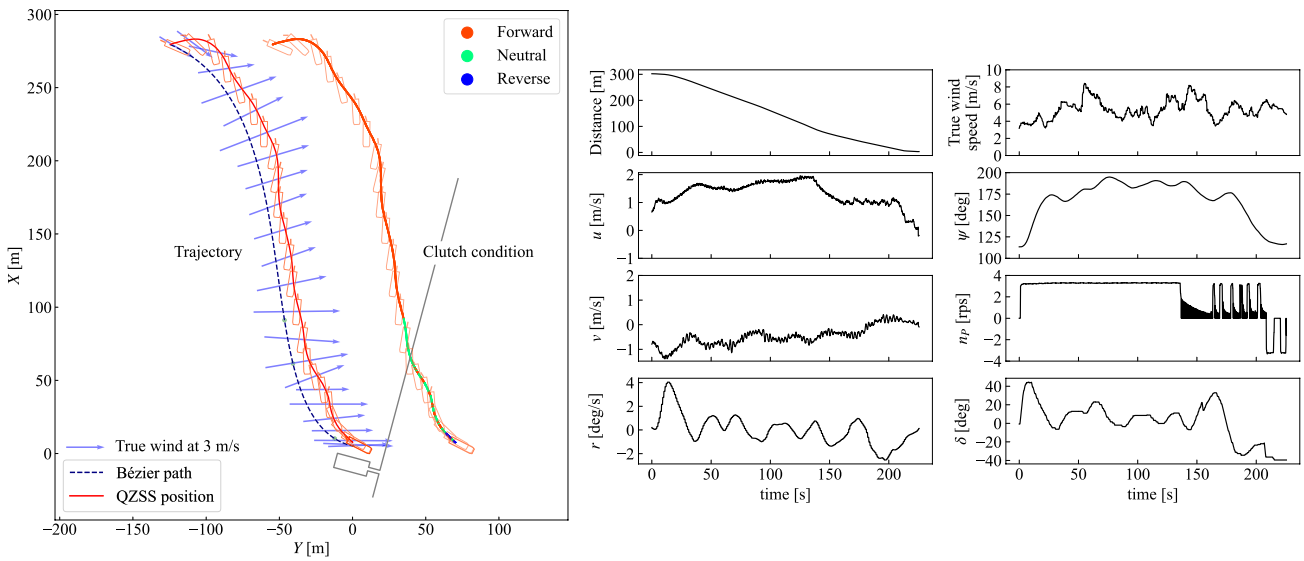

Fig. 23 Trajectory and measured values of automatic berthing control using the previous method (PPA) in wind disturbance environment

Table 7 Final state of automatic berthing control of FFPPA and PPA

Algorithm	X [m]	Y [m]	Heading angle [deg]	Surge vel. [m/s]
FFPPA	1.01	2.00	102.1 (- 1.9)	0.046 (< 0.1)
PPA	0.180	6.15	120.4 (+16.4)	0.463 (> 0.1)

the path deviation once generated could not be reduced at this wind speed without feedforward control for wind disturbance. The final states of automatic berthing control of both cases are shown in Table 7. In the example of the PPA algorithm, the final midship position satisfied the terminal condition but the final heading angle deviated significantly from the target berthing angle and did not satisfy the terminal condition. On the contrary, the FFPPA algorithm satisfied all the terminal conditions.

To evaluate the effect of the proposed method in suppressing path deviation, the deviation from the path during path following in the full-scale experiments using FFPPA and PPA is shown in Fig. 24. Here, the absolute value of cross track error is defined as deviation from the path for ease of comparison. Regarding the wind conditions during the experiments, the average true wind speeds ranged from 4.6 to 6.5 [m/s] for cases of the PPA algorithm and from 6.4 to 8.9 [m/s] for cases of the FFPPA algorithm. In both cases, no runway was set in path planning and the initial path deviation was approximately the same. However, the results with the PPA algorithm show that the initial deviation from the path persists until the end of control, indicating that the ship was approaching the floating pier at a more perpendicular angle to the original berthing path. In contrast, the results of the FFPPA algorithm show a decrease in the deviation

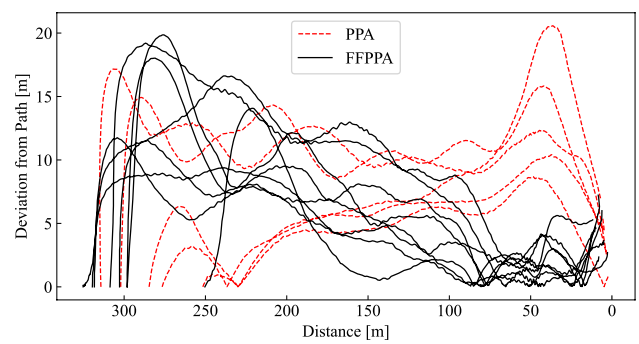

Fig. 24 Cross track error during automatic berthing control. Average true wind speed is 4.6–6.5 [m/s] for the cases with PPA (previous method, red dashed lines) and 6.4–8.9 [m//s] for the cases with FFPPA (proposed method, black solid lines)

from the path over time, indicating the effectiveness of the feedforward controller under wind disturbances.

5.4 Evaluation of AutoBerth PLC

The results of an actual ship experiment with the AutoBerth PLC for automatic berthing are shown in Fig. 25. Even when controlled by the PLC, the automatic berthing control is as accurate as that done by the laptop. In this experiment, runway was introduced in path planning, and it can be seen that the cross track error immediately after the start of control is eliminated compared to the results of previous experiments without runway of a berthing path. The cross track error at a early stage of control using the experimental ship is shown in Fig. 26. Noted that wind conditions and controllers are different, but the introduction of runway has been shown to suppress initial cross track error.

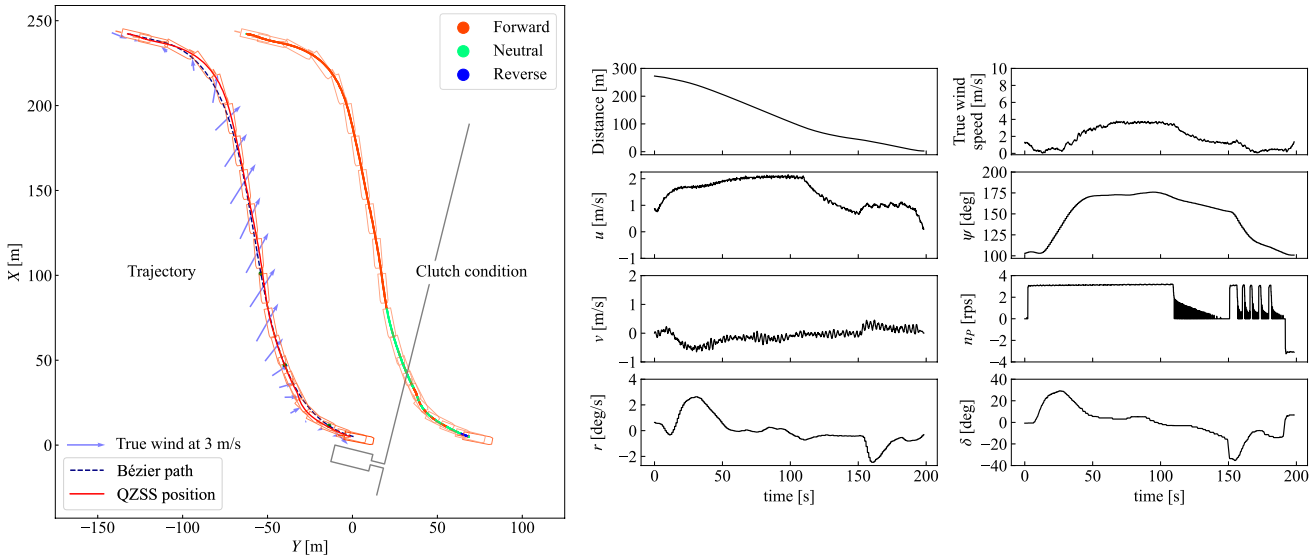

Fig. 25 Trajectory and measured values of automatic berthing control with AutoBerth PLC in wind disturbance environment

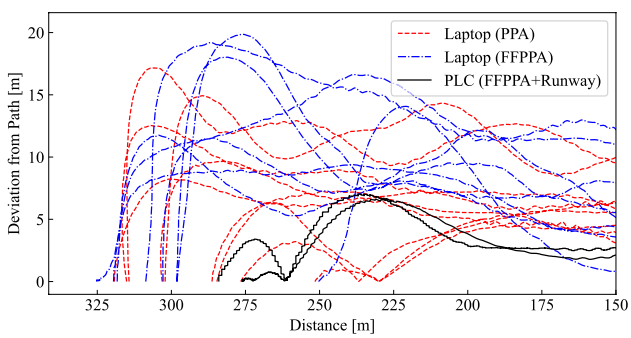

Fig. 26 Cross track error of early stage of automatic berthing control with/without straight runway of paths

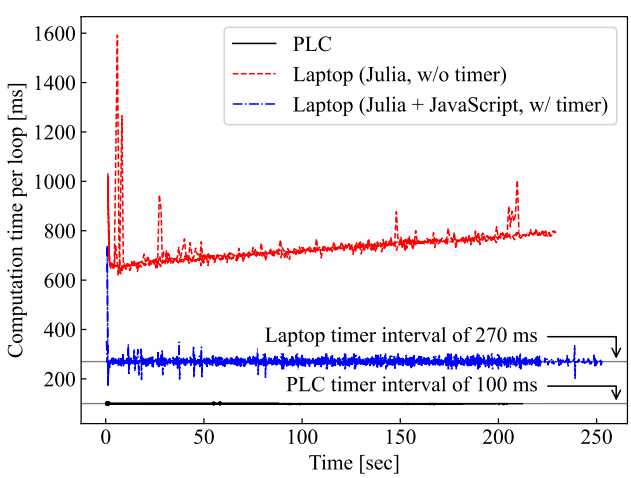

Fig. 27 Comparison of computation time per loop. Each case includes measurements of three trials respectively

The introduction of the PLC also improved stability of the control sequence; Fig. 27 compares the computation time of the control loop for the automatic berthing with two PC implementations and the AutoBerth PLC. For the implementation running on the laptop, the computation time increased with time, mainly due to the visualization process, for the implementation running sequentially without control by a timer (red dotted line). The introduction of a timer stabilized the control interval on the laptop, but it could not be less than 270 [ms] due to restrictions on the serial communication interval. Even with the timer, there were still noise in the control interval in the execution on the laptop, and there were moments when the control interval drops to nearly 800 [ms]. The AutoBerth PLC is connected via Ethernet to the main PLC system on board the ship, and the control period is controlled by a timer. The AutoBerth PLC is connected via Ethernet to the main PLC system onboard, allowing for the shortest computation interval and the most stable performance.

6 Discussion

The proposed automatic berthing control in this study mainly achieves robust control against a wind disturbance by devising rudder operation. However, the main engine speed is fixed at idling for safety reasons in the experiment, and the control of the ship's speed is limited to clutch operation only. In terms of control engineering, increasing the engine speed would create a control margin and improve robustness against disturbances of the controller. In actual ship operation, the main engine speed is increased to control the

heading angle when maneuvering for berthing in windy conditions. To realize an automatic berthing controller with performance closer to that of a human operator, it is necessary to consider control at higher engine speeds, which requires not only controller design but also adequate safety measures.

In this study, the response to deviations from the ODD is to warn operators through on-screen displays and audio alerts. In the future, if the number of personnel onboard is small, or if a fully automated vessel is envisioned, a minimum risk maneuver (MRM) should be taken when the system deviates from its ODD and the system should make effort to let a ship safe condition. Sensor integrity monitoring, perception of surrounding environments including port geometry and ships are also required for safety of MASS operation. The proposed automatic berthing system was designed to make it easier for the ship operator to make the decision to stop the automatic berthing control by designing approach angle corrections for the pure pursuit controller and approach angles of path planning, as discussed in the previous paper [2]. For example, in cases where a vessel approaches the pier at a steep angle, or even at a shallow angle, but keeps the course to the pier until the very last minute and then makes a sharp turn just before the pier, this can cause anxiety to the operator. Such operations are also unsuitable to design MRM. It is important to design a controller with enough margin to execute MRM in an emergency.

The proposed system expects the user operating the automatic berthing control to verify that the automatically generated berthing path is appropriate. In the future, when implementing automatic berthing control at higher levels of MASS, it will be necessary to automatically plan berthing paths to suit the surrounding environment. To achieve these goals, a highly accurate map of the area around the pier and depth of water information as static information and a detection system for other vessels and obstacles as dynamic information are required.

7 Conclusions

This paper proposed a practical algorithm for automatic berthing and its implementation. We showed a new simple and fast control method dealing with a wind disturbance on a path following control of automatic berthing. We analyzed the cause of cross tracking error that occurs during path following control and designed a 2-DoF controller that cancels out the lateral forces caused by the rudder and the wind load. Numerical simulations and real ship experiments were conducted to compare the proposed method with prior methods and to demonstrate the effectiveness of the proposed method. Numerical simulations were verified for various combinations of wind directions and wind speeds, and the proposed method was able to satisfy the terminal conditions of berthing for more combinations of wind conditions. In the actual ship experiment, the two methods were tested under similar wind conditions, and only the proposed method satisfied the terminal conditions and successfully berthed the ship. The experiments also showed that the proposed method was able to reduce the deviation from the path under wind disturbance. In addition, the introduction of the runway also reduced the deviation from the path during the early stages of control, which had previously occurred in path following.

To make the algorithm verified by experimentation into a more practical system, the PLC implementation was also performed. The ODD was designed and an alert function for a fallback response was implemented, along with self-diagnostic and communication monitoring functions of the PLC.

For future work, we would like to examine environmental effects other than winds, such as waves and currents, and water depth, to develop a practical ship motion model for automatic control. Research in this direction will lead to the proposal of an automatic berthing system that can be safely operated in a more practical ODD. In the process of these future studies, it is also necessary to design the ODD that enables safe operation of the automatic berthing system and to consider safety measures. These should be conducted in conjunction with the verification of the performance of an automatic control system, and it is important to identify failure scenarios based on risk assessment.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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